

CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 2 of 10)

Based on the official CBSE 2026-27 blueprint and the most heavily repeated question frameworks from previous year papers (PYQs), here is Set 2 of the predicted Class 12 Maths Half-Yearly paper. This set rigorously tests the first 80% of the syllabus, specifically excluding Linear Programming and Probability, while keeping every single question brand-new and high-probability.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x^3+5}{2}$. Find the inverse function $f^{-1}(x)$. * **Chapter:** Relations and Functions * **Confidence:** High

Q2. Find the principal value of $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence:** Very High

Q3. If $A = \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix}$ and $A^2 = I$ (where I is the identity matrix of order 2), write the relationship between α, β , and γ . * **Chapter:** Matrices * **Confidence:** Very High

Q4. If A is a non-singular square matrix of order 3 and $|A| = -4$, find the value of the determinant $|-2A|$. * **Chapter:** Determinants * **Confidence:** High

Q5. What is the derivative of $y = \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Confidence:** Very High

Q6. The parametric equations of a curve are $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$. What is the slope of the normal to this curve at $t = 2$? * **Chapter:** Application of Derivatives * **Confidence:** Medium

Q7. Evaluate the indefinite integral: $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx$. * **Chapter:** Integrals * **Confidence:** High

Q8. Evaluate the definite integral: $\int_{-\pi/4}^{\pi/4} x^3 \sin^2 x dx$. * **Chapter:** Integrals * **Confidence:** Very High

Q9. Write the integrating factor (I.F.) for the linear differential equation: $\frac{dy}{dx} - y \tan x = -\sec^2 x$. * **Chapter:** Differential Equations * **Confidence:** High

Q10. State the order and degree of the differential equation: $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$. * **Chapter:** Differential Equations * **Confidence:** Very High

Q11. Find the magnitude of the vector $\vec{a} = 3\hat{i} - 4\hat{j} + 12\hat{k}$. * **Chapter:** Vector Algebra * **Confidence:** High

Q12. If $|\vec{a}| = 5$, $|\vec{b}| = 4$, and their scalar dot product $\vec{a} \cdot \vec{b} = 10$, find the magnitude of their cross product $|\vec{a} \times \vec{b}|$. * **Chapter:** Vector Algebra * **Confidence:** Very High

Q13. Find the direction cosines of the straight line passing through the points $(2, -1, 4)$ and $(1, 1, -2)$. * **Chapter:** Three-Dimensional Geometry * **Confidence:** High

Q14. Find the perpendicular distance of the plane $2x - 3y + 6z = 14$ from the origin. * **Chapter:** Three-Dimensional Geometry * **Confidence:** High

Q15. If a square matrix A is both a symmetric matrix and a skew-symmetric matrix, what type of matrix must A be? * **Chapter:** Matrices * **Confidence:** Medium

Q16. Let $A = \{1, 2, 3\}$. How many equivalence relations can be defined on set A containing the elements $(1, 2)$ and $(2, 1)$? * **Chapter:** Relations and Functions * **Confidence:** High

Q17. What is the rate of change of the volume of a sphere with respect to its surface area when the radius is 2 cm? * **Chapter:** Application of Derivatives * **Confidence:** Very High

Q18. Evaluate the indefinite integral: $\int \frac{1}{\sin^2 x \cos^2 x} dx$. * **Chapter:** Integrals * **Confidence:** Very High

Q19. (Assertion-Reason): Assertion (A): The greatest integer function $f(x) = [x]$ is not continuous at $x = 1$. **Reason (R):** The left-hand limit and right-hand limit of $f(x)$ at $x = 1$ exist but are not equal to each other. Choose the correct option. * **Chapter:** Continuity and Differentiability * **Confidence:** Very High

Q20. (Assertion-Reason): Assertion (A): The vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear. **Reason (R):** Two non-zero vectors are collinear if their cross product is the zero vector. Choose the correct option. * **Chapter:** Vector Algebra * **Confidence:** High

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Prove algebraically that $\sin^{-1}\left(\frac{8}{17}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(\frac{77}{85}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence:** High

Q22. If $A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$, find A^{-1} using the determinant and adjoint formula. * **Chapter:** Matrices and Determinants * **Confidence:** High

Q23. If $x = a(\cos \theta + \theta \sin \theta)$ and $y = a(\sin \theta - \theta \cos \theta)$, find $\frac{dy}{dx}$ in its simplest form. * **Chapter:** Continuity and Differentiability * **Confidence:** Very High

Q24. Solve the differential equation by the method of separation of variables: $\frac{dy}{dx} = \frac{x+1}{2-y}$. * **Chapter:** Differential Equations * **Confidence:** Medium

Q25. Find a unit vector strictly perpendicular to the plane containing the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$. * **Chapter:** Vector Algebra * **Confidence:** Very High

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R on the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b^2\}$ is neither reflexive, nor symmetric, nor transitive. * **Chapter:** Relations and Functions * **Confidence:** Very High

Q27. Using the standard properties of determinants, prove that:
$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$
. *

Chapter: Determinants * **Confidence:** Very High

Q28. Differentiate the function $\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$ with respect to x , assuming x lies in the principal domain. * **Chapter:** Continuity and Differentiability * **Confidence:** High

Q29. Evaluate the definite integral using limit properties: $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$. * **Chapter:** Integrals * **Confidence:** Very High

Q30. Find the exact intervals in which the function $f(x) = \sin 3x$, where $x \in [0, \frac{\pi}{2}]$, is strictly increasing or strictly decreasing. * **Chapter:** Application of Derivatives * **Confidence:** High

Q31. Find the Cartesian and vector equations of the line passing through the point $(-1, 4, -2)$ and parallel to the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. * **Chapter:** Three-Dimensional Geometry * **Confidence:** High

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Use the matrix inverse method to solve the following system of linear equations: $x - y + z = 4$, $2x + y - 3z = 0$, $x + y + z = 2$. * **Chapter:** Matrices and Determinants * **Confidence:** Very High

Q33. Prove mathematically that the total surface area of a solid right circular cylinder of a given (fixed) volume is minimum when its height is exactly equal to the diameter of its base. * **Chapter:** Application of Derivatives * **Confidence:** Very High

Q34. Evaluate the following indefinite integral using standard algebraic transformations: $\int \frac{x^2+1}{x^4+1} dx$. * **Chapter:** Integrals * **Confidence:** High

Q35. Using the method of integration, find the exact area of the region common to (bounded by) the two circles $x^2 + y^2 = 1$ and $(x-1)^2 + y^2 = 1$. * **Chapter:** Application of Integrals * **Confidence:** Very High

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Relations and Functions (School Database System) In a school's digital administration system, student data is managed using mathematical mappings. Let A be the set of all students currently enrolled in the school. The database uses a function $f: A \rightarrow \mathbb{N}$ to assign each student their unique admission roll number. Additionally, let $B = \{\text{Science, Commerce, Arts}\}$ represent the three available streams. Another function $g: A \rightarrow B$ assigns each student to their chosen stream. Assume that every stream has at least one student enrolled. (i) Is the function f one-one (injective)? Give a brief mathematical reason. (1 Mark) (ii) Based on the information provided, is the function g onto (surjective)? Give a reason. (1 Mark) (iii) A relation R is defined on set A such that $(x, y) \in R$ if and only if student x and student y belong to the exact same stream. Prove that R is an equivalence relation. (2 Marks) * **Chapter:** Relations and Functions * **Confidence:** High

Q37. Case Study 2: Differential Equations (Newton's Law of Cooling) In a physics laboratory, an experiment is conducted to verify Newton's Law of Cooling, which states that the rate of change of temperature of an object is directly proportional to the difference between its own temperature and the ambient (surrounding) temperature. A beaker of hot liquid is cooling in a room where the ambient temperature is constantly maintained at 20°C . Let T represent the temperature of the liquid at time t minutes, and k be a positive constant of proportionality. (i) Formulate the first-order differential equation representing this cooling process. (1 Mark) (ii) Solve the differential equation to find the general solution expressing T as a function of t . (1 Mark) (iii) If the initial temperature of the liquid was 100°C and it drops to 60°C in exactly 10 minutes, what will be the exact temperature of the liquid after another 10 minutes (i.e., at $t = 20$)? (2 Marks) * **Chapter:** Differential Equations * **Confidence:** Very High

Q38. Case Study 3: 3D Geometry (Bridge Architecture) Civil engineers are using 3D coordinate geometry to model a suspension bridge. Two main diagonal support cables are mathematically represented by straight lines in 3D space. Cable 1 lies along the vector equation: $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(3\hat{i} + \hat{j} - \hat{k})$. Cable 2 lies along the vector equation: $\vec{r} = (\hat{i} + \hat{j} + 2\hat{k}) + \mu(-\hat{i} + 2\hat{j} + \hat{k})$. To ensure maximum stability, the architects need to determine the shortest mathematical distance between these two cables to place a horizontal connecting rod. (i) Write the direction vectors $\vec{b}_1$ and $\vec{b}_2$ of Cable 1 and Cable 2. (1 Mark) (ii) Find the cross product $(\vec{b}_1 \times \vec{b}_2)$ of the two direction vectors. (1 Mark) (iii) Calculate the exact shortest distance between the two cables. Based on the result, state whether the cables intersect or are skew lines. (2 Marks) * **Chapter:** Three-Dimensional Geometry * **Confidence:** High

Answer Key

Q1. $y = \frac{x^3+5}{2} \Rightarrow 2y - 5 = x^3 \Rightarrow x = \sqrt[3]{2y-5}$. Therefore, $f^{-1}(x) = \sqrt[3]{2x-5}$. **Q2.** $\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = \pi - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$. **Q3.** $A^2 = \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} = \begin{pmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \alpha^2 + \beta\gamma \end{pmatrix}$. For $A^2 = I$, $\alpha^2 + \beta\gamma = 1$. **Q4.** $|-2A| = (-2)^3|A| = -8 \times (-4) = 32$. **Q5.** $y = \tan^{-1}\left(\frac{2\sin(x/2)\cos(x/2)}{2\cos^2(x/2)}\right) = \tan^{-1}(\tan(x/2)) = \frac{x}{2}$. Thus, $\frac{dy}{dx} = \frac{1}{2}$. **Q6.** $\frac{dx}{dt} = 2t + 3$, $\frac{dy}{dt} = 4t - 2$. At $t = 2$, $\frac{dx}{dt} = 7$, $\frac{dy}{dt} = 6$. Slope of tangent = $\frac{6}{7}$. Slope of normal = $-\frac{7}{6}$. **Q7.** Let $\tan x = t \Rightarrow \sec^2 x dx = dt$. Integral becomes $\int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + C = \sin^{-1}(\tan x) + C$. **Q8.** Let $f(x) = x^3 \sin^2 x$. Since $f(-x) = (-x)^3 \sin^2(-x) = -x^3 \sin^2 x = -f(x)$, it is an odd function. Definite integral is 0. **Q9.** $P = -\tan x$. I.F. $= e^{\int -\tan x dx} = e^{\log(\cos x)} = \cos x$. **Q10.** Squaring both sides gives $\left(\frac{d^2y}{dx^2}\right)^2 = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$. Order = 2, Degree = 2. **Q11.** Magnitude = $\sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. **Q12.** $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 \Rightarrow |\vec{a} \times \vec{b}|^2 + 10^2 = 5^2 \times 4^2 \Rightarrow |\vec{a} \times \vec{b}|^2 + 100 = 400 \Rightarrow |\vec{a} \times \vec{b}| = \sqrt{300} = 10\sqrt{3}$. **Q13.** DRs: $1 - 2, 1 - (-1), -2 - 4 \Rightarrow -1, 2, -6$. Magnitude = $\sqrt{1 + 4 + 36} = \sqrt{41}$. DCs are $\frac{-1}{\sqrt{41}}, \frac{2}{\sqrt{41}}, \frac{-6}{\sqrt{41}}$. **Q14.** Distance $d = \frac{|-14|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{14}{\sqrt{49}} = \frac{14}{7} = 2$ units. **Q15.** Since $A = A^T$ (symmetric) and $A = -A^T$ (skew-symmetric), we have $A = -A \Rightarrow 2A = O$. Thus, A is a null (zero) matrix. **Q16.** 2 relations. The smallest is $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$, and the largest is the universal relation $A \times A$. **Q17.** $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2$. $S = 4\pi r^2 \Rightarrow \frac{dS}{dr} = 8\pi r$. $\frac{dV}{dS} = \frac{4\pi r^2}{8\pi r} = \frac{r}{2}$. At $r = 2$, rate is 1 cm. **Q18.** Numerator $1 = \sin^2 x + \cos^2 x$. Separating denominator gives $\int (\sec^2 x + \operatorname{cosec}^2 x) dx = \tan x - \cot x + C$. **Q19.** Option A (Both A and R are true, and R is the correct explanation of A. LHL is 0, RHL is 1). **Q20.** Option A (Both A and R are true, and R is the correct explanation. $\vec{b} = -2\vec{a}$, meaning they are scalar multiples / parallel). **Q21.** Let $\alpha = \sin^{-1}\left(\frac{8}{17}\right) \Rightarrow \sin \alpha = \frac{8}{17}, \cos \alpha = \frac{15}{17}$. Let $\beta = \sin^{-1}\left(\frac{3}{5}\right) \Rightarrow \sin \beta = \frac{3}{5}, \cos \beta = \frac{4}{5}$. Using $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta = \left(\frac{8}{17}\right)\left(\frac{4}{5}\right) + \left(\frac{15}{17}\right)\left(\frac{3}{5}\right) = \frac{32+45}{85} = \frac{77}{85}$. Proved. **Q22.** $|A| = 1(1) - (-2)(2) = 5$. Adjoint matrix is found by swapping principal diagonals and negating non-principal: $\operatorname{adj} A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$. $A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$. **Q23.** $\frac{dz}{d\theta} = a(-\sin \theta + \sin \theta + \theta \cos \theta) = a\theta \cos \theta$. $\frac{dy}{d\theta} = a(\cos \theta - (\cos \theta - \theta \sin \theta)) = a\theta \sin \theta$. $\frac{dy}{dx} = \frac{a\theta \sin \theta}{a\theta \cos \theta} = \tan \theta$. **Q24.** $(2 - y)dy = (x + 1)dx$. Integrating both sides: $2y - \frac{y^2}{2} = \frac{x^2}{2} + x + C$. **Q25.** Cross product $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 3 & -1 \end{vmatrix} = \hat{i}(1 - 6) - \hat{j}(-1 - 4) + \hat{k}(3 + 2) = -5\hat{i} + 5\hat{j} + 5\hat{k}$. Magnitude = $\sqrt{25 + 25 + 25} = 5\sqrt{3}$. Unit vector = $\frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$.

Q26. Not reflexive: $\frac{1}{2} \not\leq (\frac{1}{2})^2 = \frac{1}{4}$. Not symmetric: $-1 \leq 2^2 = 4$ but $2 \not\leq (-1)^2 = 1$. Not transitive: $2 \leq (-1.5)^2 = 2.25$ and $-1.5 \leq 1^2 = 1$, but $2 \not\leq 1^2 = 1$. **Q27.** Take a, b, c common from rows R_1, R_2, R_3 respectively. Det becomes

$$abc \begin{vmatrix} 1/a + 1 & 1/a & 1/a \\ 1/b & 1/b + 1 & 1/b \\ 1/c & 1/c & 1/c + 1 \end{vmatrix}. \text{ Apply } R_1 \rightarrow R_1 + R_2 + R_3 \text{ to get common factor } (1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \text{ in Row 1. Expand}$$

to show remaining determinant simplifies exactly to 1. **Q28.**

$$\frac{\cos x}{1 + \sin x} = \frac{\cos^2(x/2) - \sin^2(x/2)}{(\cos(x/2) + \sin(x/2))^2} = \frac{\cos(x/2) - \sin(x/2)}{\cos(x/2) + \sin(x/2)} = \frac{1 - \tan(x/2)}{1 + \tan(x/2)} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right). \tan^{-1}(\tan(\dots)) = \frac{\pi}{4} - \frac{x}{2}. \text{ Derivative}$$

$$= -\frac{1}{2}. \text{ Q29. Apply } \int_0^a f(x) dx = \int_0^a f(a-x) dx. I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx. \text{ Adding the original and transformed integrals:}$$

$$2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2} \implies I = \frac{\pi}{4}. \text{ Q30. } f'(x) = 3 \cos 3x. \text{ For strictly increasing:}$$

$$\cos 3x > 0 \implies 3x \in (0, \frac{\pi}{2}) \implies x \in [0, \frac{\pi}{6}). \text{ For strictly decreasing:}$$

$$\cos 3x < 0 \implies 3x \in (\frac{\pi}{2}, \frac{3\pi}{2}) \implies x \in (\frac{\pi}{6}, \frac{\pi}{2}]. \text{ Q31. Parallel lines share identical direction ratios. Using point}$$

$$(-1, 4, -2) \text{ and DRs } 2, 3, 4: \text{ Cartesian form is } \frac{x+1}{2} = \frac{y-4}{3} = \frac{z+2}{4}. \text{ Vector form is } \vec{r} = (-\hat{i} + 4\hat{j} - 2\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}).$$

$$\text{Q32. Let } AX = B. |A| = \begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{vmatrix} = 1(4) - (-1)(5) + 1(1) = 10 \neq 0. \text{ Find cofactors to form adjoint.}$$

$$A^{-1} = \frac{1}{10} \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}. X = A^{-1}B \implies x = 2, y = -1, z = 1. \text{ Q33. Volume } V = \pi r^2 h \implies h = \frac{V}{\pi r^2}. \text{ Surface}$$

$$\text{Area } S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{2V}{r}. \text{ Differentiate wrt } r: \frac{dS}{dr} = 4\pi r - \frac{2V}{r^2} = 0 \implies V = 2\pi r^3. \text{ Substitute } V \text{ back:}$$

$$\pi r^2 h = 2\pi r^3 \implies h = 2r \text{ (height = diameter). Second derivative } \frac{d^2 S}{dr^2} = 4\pi + \frac{4V}{r^3} > 0 \text{ validates minimum area. Q34.}$$

$$\text{Divide numerator and denominator by } x^2: \int \frac{1+1/x^2}{x^2+1/x^2} dx = \int \frac{1+1/\frac{dx}{dt}}{(x-1/x)^2+2} dx. \text{ Substitute } t = x - \frac{1}{x} \implies dt = (1 + \frac{1}{x^2}) dx.$$

$$\text{Integral becomes } \int \frac{dt}{t^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t}{\sqrt{2}}\right) + C = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) + C. \text{ Q35. The circles intersect at } x = \frac{1}{2}. \text{ By}$$

$$\text{symmetry, Area} = 2 \left[\int_0^{1/2} \sqrt{1 - (x-1)^2} dx + \int_{1/2}^1 \sqrt{1 - x^2} dx \right]. \text{ Standard evaluation of the limits gives } \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \text{ sq units.}$$

Q36. (i) Yes, f is one-one because every student receives a distinct, unique admission roll number. No two students map to the same roll number. (ii) Yes, g is onto. Since every stream has at least one student enrolled, every element in the codomain B has at least one pre-image in domain A . (iii) Reflexive: Student x is in the same stream as themselves. Symmetric: If x is in the same stream as y , then y is in the same stream as x . Transitive: If x, y share a stream, and y, z share a stream, then x, z share a stream. Hence, it is an equivalence relation. **Q37.** (i) $\frac{dT}{dt} = -k(T - 20)$. (ii) Separating variables $\frac{dT}{T-20} = -k dt$ yields $\ln |T - 20| = -kt + C$, so the general solution is $T(t) = 20 + Ae^{-kt}$. (iii) At $t = 0, T = 100 \implies A = 80$. At $t = 10, T = 60 \implies 60 = 20 + 80e^{-10k} \implies e^{-10k} = \frac{1}{2}$. At $t = 20, T(20) = 20 + 80(e^{-10k})^2 = 20 + 80(\frac{1}{4}) = 40^\circ\text{C}$.

$$\text{Q38. (i) Cable 1: } \vec{b}_1 = 3\hat{i} + \hat{j} - \hat{k}. \text{ Cable 2: } \vec{b}_2 = -\hat{i} + 2\hat{j} + \hat{k}. \text{ (ii) Cross product } \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -1 \\ -1 & 2 & 1 \end{vmatrix} = 3\hat{i} - 2\hat{j} + 7\hat{k}.$$

$$\text{(iii) Displacement vector } \vec{a}_2 - \vec{a}_1 = (1 - 2)\hat{i} + (1 - (-1))\hat{j} + (2 - 1)\hat{k} = -\hat{i} + 2\hat{j} + \hat{k}. \text{ Since}$$

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = (-1)(3) + (2)(-2) + (1)(7) = -3 - 4 + 7 = 0. \text{ Shortest distance is } 0, \text{ which proves the cables naturally intersect (are not skew).}$$